ETH zürich

KOF

Principles of Macroeconomics

Lecture 1: Introduction

Prof. Dr. Jan-Egbert Sturm

MTEC's Vision is…

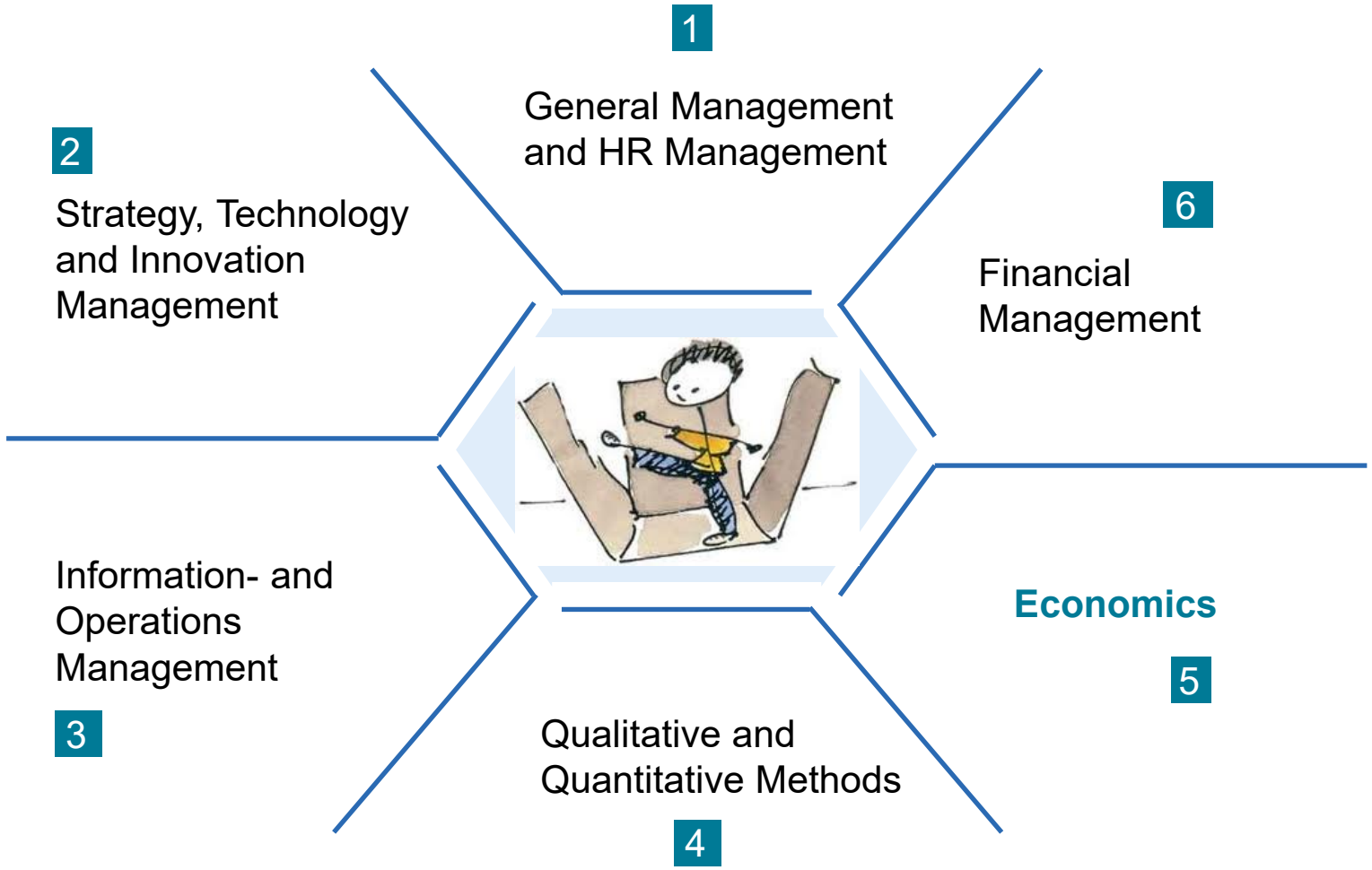
... to understand, shape and relate the interactions between technology, organizations and society and their impact on the sustainable use of natural and human resources.

Learning objectives

- We aim to develop MTEC Students to ...
  - … understand Technology, Management, AND Economics
  - … enhance their analytical skills to become systemic problem-solver based on a solid scientific background
  - … develop decision making and leadership skills in an interdisciplinary and intercultural environment

- The ability to work in teams is at the core of innovation, and we will learn it in practice thanks to the multicultural and interdisciplinary context of the class.

D-MTEC skills areas (Kompetenzfelder)



Minimum requirements for writing a Master Thesis in **Economics** at D-MTEC

- **Master of Science**
  - Basic requirements at all chairs in economics
    - *Principles of Microeconomics* & *Decisions and Markets*
    - *Principles of Macroeconomics* & *Economic Growth, Cycles and Policy*
  - Additional requirements (specific to each chair)
    - In case of the Chair of Applied Macroeconomics
      - *Principles of Econometrics*
      - (*Intermediate Econometrics* is highly recommended)
- **Master of Advanced Studies** (MAS)
  - Basic requirements at all chairs
    - *Principles of Microeconomics* & *Principles of Macroeconomics*
    - 6 Credits in Economics
  - Additional requirements (specific to each chair)
    - In case of the Chair of Applied Macroeconomics: none

General information (1)

- Course information “363-0565-00L Principles of Macroeconomics HS2025”
  - https://moodle-app2.let.ethz.ch/course/view.php?id=25353 (Login with your ETHZ credentials)
  - Problem sets will be made available on Moodle

- Literature
  - **N. Gregory Mankiw and Mark P. Taylor (2023): *Economics*, 6th Edition, Cengage Learning**
    - Same book can be used in “363-0503-00L Principles of Microeconomics”
  - Please follow the instructions in this **file** and visit www.cengage.uk to apply a 25% discount code

ETH EduApp

- The **ETH EduApp** is a mobile and web application to help students through everyday university life (personal timetables, ETH maps, directions), as well as to foster the interaction during courses (clicker questions, feedback, course channels).
  - **Clicker questions** allow you to interact and answer the lecturer's multiple-choice questions during the lecture
  - **Course channel** allows you to provide opinions, suggestions, questions or ideas which cannot be provided in high-attendance lectures. It allows to note each other's concerns and the feedback you provide can contribute to the ongoing improvement of course quality.
  - **Semester Feedback** function allows students to provide their final feedback to the course.
- More information: https://ethz.ch/staffnet/en/teaching/academic-support/it-services-teaching/teaching-applications/eduapp-service.html

General Information (2)

- This lecture is being recorded for educational purposes
  - We will make these recordings available through Moodle for students registered for this course
    - (Sometimes this might take a while – depending on zoom and/or multimedia services)
    - Recordings will be through the Multimedia Services of the ETH Zurich
      - Login with your ETHZ credentials

- Prof. Dr. Jan-Egbert Sturm
  - office hours: on appointment
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- Roxane Spitznagel / Aline Scheurer
  - office hours: on appointment
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General Information (3)

- Exam date:
  - Likely date (but no warranty!): Tuesday, 13.01.2026, 14:00-15:30
  - Rooms: will be announced

- Type of exam:
  - 90 minutes written paper exam
  - Closed book
  - **Everyone** who wants to get credits will have to take the exam.
    There are no “Sitzscheine”.

General Information (4)

- Exam registration
  - Please register for the exam on myStudies (3rd and 4th of week of each semester).
  - Enrolment in the course is NOT equal to an exam registration.
  - Further information:

    https://ethz.ch/applications/teaching/en/applications/mystudies/matriculation/examinations.html

    https://ethz.ch/students/en/studies/performance-assessments/registration.html

- Retake/repetition Exam
  - Only for students that failed in the final exam
    (for special cases: please check with the examination office)
    - Likely date (but no warranty!): Monday, 23.02.2026, 14:00-15:30

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Karl Brunner Distinguished Lecture

"Inflation" by John H. Cochrane,
Hoover Institution (Stanford University)

Thursday, 2 October 2025, ETH Zurich, Audi Max (HG F 30)
Drinks reception: 4.45 – 5.30 pm; Lecture: 5.30 – 7.00 pm

Welcome by Joël Mesot
President of ETH Zurich

Introduction by Martin Schlegel
Chairman of the Governing Board, Swiss National Bank

Moderated by Tanya König

Live streaming www.snb.ch, Research TV

Please note that there will be a security check at the entrance to the lecture hall. Please arrive well in advance of the start of the event and refrain from bringing bags that are larger than A4 (21 x 30 cm).

Hosted by **KOF** Swiss Economic Institute and Chair of Macroeconomics: Innovation and Policy

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«Welt aus den Fugen – Stresstest für die Schweiz? Risiken erkennen, Resilienz stärken, Zukunft sichern»

KOF Prognosetagung
25. September 2025, ab 12:30 Uhr

ExpertInnen:
Sermîn Faki, Nachrichtenchefin CH Media
Prof. Dr. Hans Gersbach, Co-Direktor KOF, ETH Zürich
Anne Schwöbel, Mitglied Verwaltungsrat, B. Braun Medical AG
Prof. Dr. Jan-Egbert Sturm, Direktor KOF, ETH Zürich

Mehr Infos

Schedule

Date	Lecture	Topic	Chapters
16/09/2025	1	Introduction	1, 2
23/09/2025	2	National Accounting	20
30/09/2025	3	Production and Growth	21
07/10/2025	4	Unemployment and the labour market	22
14/10/2025	5	Saving, investment and the financial system	23
21/10/2025	6	The Monetary System	24
28/10/2025	7	International Trade	3,15,17
04/11/2025	8	Open-Economy Macroeconomics	25
11/11/2025	9	Business Cycles	26
18/11/2025	10	Keynesian Economics and IS-LM Analysis	27
25/11/2025	11	Aggregate Demand and Aggregate Supply	28
02/12/2025	12	Monetary and Fiscal Policy	29
09/12/2025	13	Phillips Curve / Supply-side policies	30,31
16/12/2025	14	Exam preparation	

Principles of Economics

- A household and an economy face many decisions:
  - Who will work?
  - What goods and how many of them should be produced?
  - What resources should be used in production?
  - At what price should the goods be sold?

- Society and Scarce Resources:
  - The management of society's resources is important because resources are scarce
  - ***Scarcity***. . . means that society has limited resources and therefore cannot produce all the goods and services people wish to have

- ***Economics*** is the study of how society manages its scarce resources

10 Principles of Economics

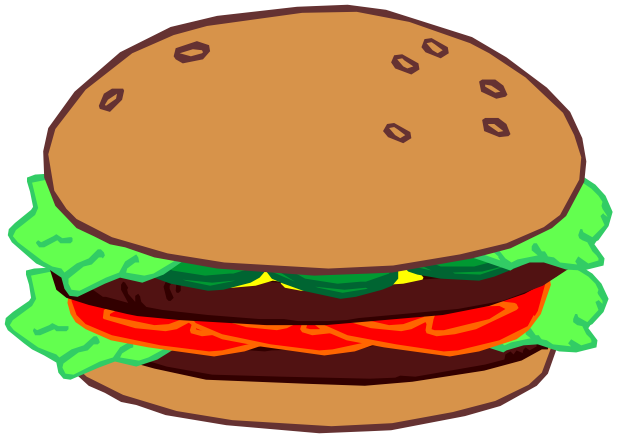
1. People face trade-offs
2. The cost of something is what you give up to get it
3. Rational people think at the margin
4. People respond to incentives
5. Trade can make everyone better off
6. Markets are usually a good way to organize economic activity
7. Governments can sometimes improve economic outcomes
8. A country's standard of living depends on its ability to produce goods and services
9. Prices rise when the government prints too much money
10. Society faces a short-run trade-off between inflation and unemployment

HOW PEOPLE MAKE DECISIONS

1. People face trade-offs
2. The cost of something is what you give up to get it
3. Rational people think at the margin
4. People respond to incentives

Principle #1: People Face Trade-offs

- “There is no such thing as a free lunch!”



- To get one thing, we usually have to give up another thing
  - Guns v. butter
  - Food v. clothing
  - Leisure time v. work
  - Efficiency v. equity

Making decisions requires trading off one goal against another.

- Efficiency v. Equity
  - *Efficiency* means society gets the most that it can from its scarce resources
  - *Equity* means the benefits of those resources are distributed fairly among the members of society

Principle #2: The Cost of Something Is What You Give Up to Get It

- Decisions require comparing costs and benefits of alternatives
  - Whether to go to college or to work?
  - Whether to study or go out on a date?
  - Whether to go to class or sleep in?

- The *opportunity cost* of an item is what you give up to obtain that item
  - It is a ratio expressed as the sacrifice in one good in terms of the gain in the other

Principle #3: Rational Economic Agents Think at the Margin

- An *economic agent* is an individual, firm or organization that has an impact in some way on an economy
- Being *rational* is the assumption that economic agents can make consistent choices between alternatives
- *Marginal changes* are small, incremental adjustments to an existing plan of action

People make decisions by comparing costs and benefits at the margin

Principle #4: People Respond to Incentives

Assuming rational people make rational decisions, then….

- Marginal changes in costs or benefits motivate people to respond
- The decision to choose one alternative over another occurs when that alternative's marginal benefits exceed its marginal costs!
- Public policies can create incentives or disincentives that alter behaviour
- Sometimes policymakers fail to understand how policies alter incentives and behaviour

HOW PEOPLE INTERACT

5. Trade can make everyone better off
6. Markets are usually a good way to organize economic activity
7. Governments can sometimes improve economic outcomes

Principle #5: Trade Can Make Everyone Better Off

- People gain from their ability to trade with one another
- Competition results in gains from trading
- Trade allows people to specialize in what they do best

- However, sometimes this leads to people losing their jobs as production moves overseas. They need alternative skills and the area needs to attract new investment, which does not always happen fast

Principle #6: Markets Are Usually a Good Way to Organize Economic Activity

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- A *market economy* is an economy that allocates resources through the decentralized decisions of many firms and households as they interact in markets for goods and services
  - Households decide what to buy and who to work for
  - Firms decide who to hire and what to produce

- Market prices reflect both
  - the value of a product to consumers and
  - the cost of the resources used to produce it
- Adam Smith (1723-1790) made the observation that households and firms interacting in markets act as if guided by an “invisible hand”
  - Because households and firms look at prices when deciding what to buy and sell, they unknowingly take into account the social costs of their actions
  - As a result, prices guide decision makers to reach outcomes that tend to maximize the welfare of society as a whole

Market vs. Planned economic systems

- An economic system is the way in which resources are organized and allocated to provide for the needs of an economy's citizens
- Capitalist Economic System
  - A capitalist economic system incorporates the principles of the private ownership of factors of production to produce goods and services which are exchanged through a price mechanism and where production is operated primarily for profit
  - Critics of the capitalist system argue that they are inherently unstable and lurch from boom-to-bust and that owners of the factors of production are too powerful and can distort resource allocation
- Planned Economic System
  - A planned economic system is when economic activity organized by central planners who decide the answers to the fundamental economic questions
  - Most countries that once had centrally planned economies have abandoned this system and are developing more market-based economies

Principle #7: Governments Can Sometimes Improve Market Outcomes

- Markets work only if property rights are enforced
  - *Property rights* are the ability of an individual to own and exercise control over a scarce resource
- *Market failure* occurs when the market fails to allocate resources efficiently
- When the market fails (breaks down) government can intervene to promote efficiency and equity
- Market failure may be caused by:
  - an *externality*, which is the impact of one person or firm's actions on the well-being of a bystander
  - *market power,* which is the ability of a single person or firm to unduly influence market prices

Microeconomics and Macroeconomics

- *Microeconomics* focuses on the individual parts of the economy
  - How households and firms make decisions and how they interact in specific markets

- *Macroeconomics* looks at the economy as a whole
  - Economy-wide phenomena, including inflation, unemployment, and economic growth

- Changes in the overall economy arise from the decisions of millions of individuals, so macroeconomics and microeconomics are linked

HOW THE ECONOMY AS A WHOLE WORKS

8. A country's standard of living depends on its ability to produce goods and services
9. Prices rise when the government prints too much money
10. Society faces a short-run trade-off between inflation and unemployment

Principle #8: A Country's Standard of Living Depends on Its Ability to Produce Goods and Services

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- *Economic growth* - the increase in the amount of goods and services in an economy over a period of time.
- *Gross domestic product per head* - the market value of all final goods and services produced within a country in a given period of time divided by the population of a country to give a per capita figure
- *Standard of living* may be measured in different ways:
  - By comparing (inflation adjusted) personal incomes
  - By comparing the total market value of a nation's production
- Almost all variations in living standards are explained by differences in countries' productivities
- *Productivity* is the amount of goods and services produced from each hour of a worker's time

Principle #9: Prices Rise When the Government Prints Too Much Money

- *Inflation* is an increase in the overall level of prices in the economy
- One cause of inflation is the growth in the quantity of money
- When the government creates large quantities of money, the value of the money falls

- Too high and too low inflation imposes various costs on society

Principle #10: Society Faces a Short-run Trade-off between Inflation and Unemployment

- The Phillips Curve illustrates the trade-off between inflation and unemployment:
  - Lower inflation is associated with higher unemployment
    - It's a short-run trade-off!
  - The trade-off plays a key role in the analysis of the *business cycle* – fluctuations in economic activity, such as employment and production

Thinking Like an Economist

Every field of study has its own terminology

- Mathematics
  - integrals / axioms / vector spaces
- Psychology
  - ego / id / cognitive dissonance
- Law
  - promissory / estoppel / torts / venues
- Economics
  - supply / opportunity cost / elasticity / consumer surplus / demand / comparative advantage / deadweight loss

Thinking Like an Economist

Economics trains you to. . . .

- think in terms of alternative
- evaluate the cost of individual and social choices
- examine and understand how certain events and issues are related

THE ECONOMIST AS A SCIENTIST

The economic way of thinking . . .

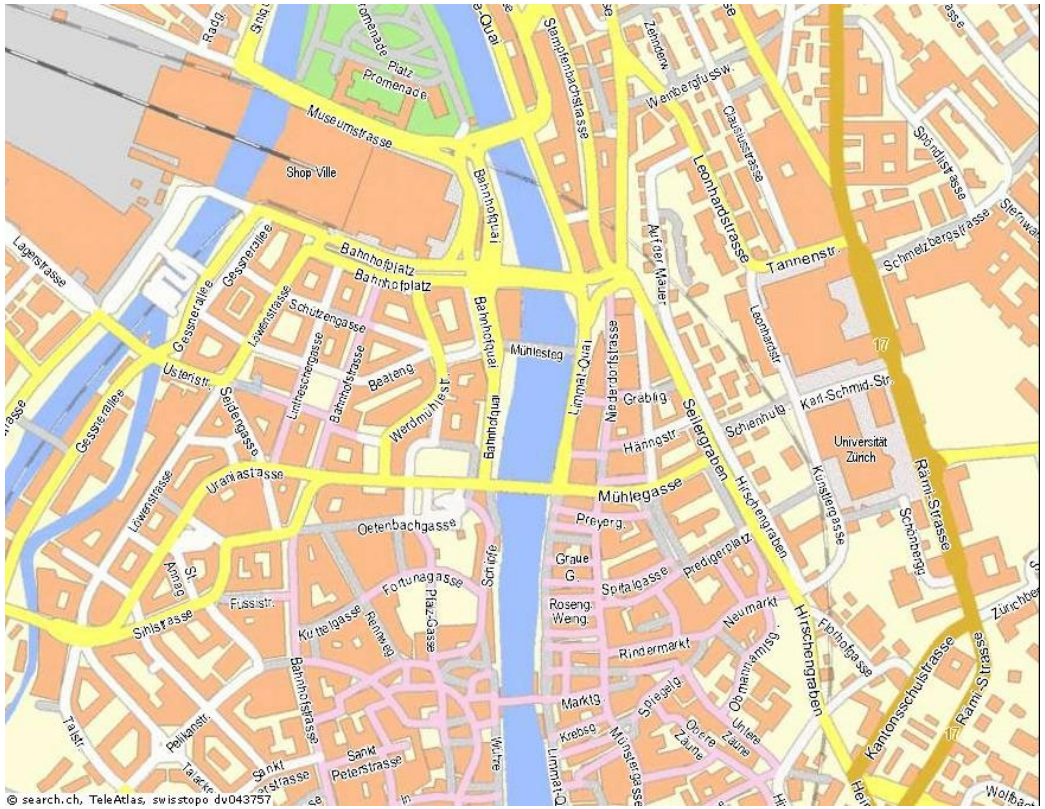
- involves thinking analytically and objectively
- makes use of the scientific method
  (Observation, Theory, and More Observation)
  - Uses abstract models to help explain how a complex, real world operates
  - Develops theories, collects and analyses data to evaluate the theories

Empiricism

- Empirical information is gathered from:
  - Observation.
  - Experience.
  - Experiment.
- An hypothesis is an assumption that we want to test.
- Two types of reasoning are used in academics
  - Inductive reasoning refers to the process of observation from which patterns might be formed that provide evidence for a hypothesis which may lead to a theory.
  - Deductive reasoning begins with a theory from which a hypothesis is drawn.
    The hypothesis is then subject to observation and either confirmation or rejection.
- Theories can be used to explain something and to make predictions.
- Things change; old theories can be rejected, and new theories come along

The Role of Assumptions

- Economists make assumptions in order to make the world easier to understand
- The art in scientific thinking is deciding which assumptions to make
- Economists use different assumptions to answer different questions



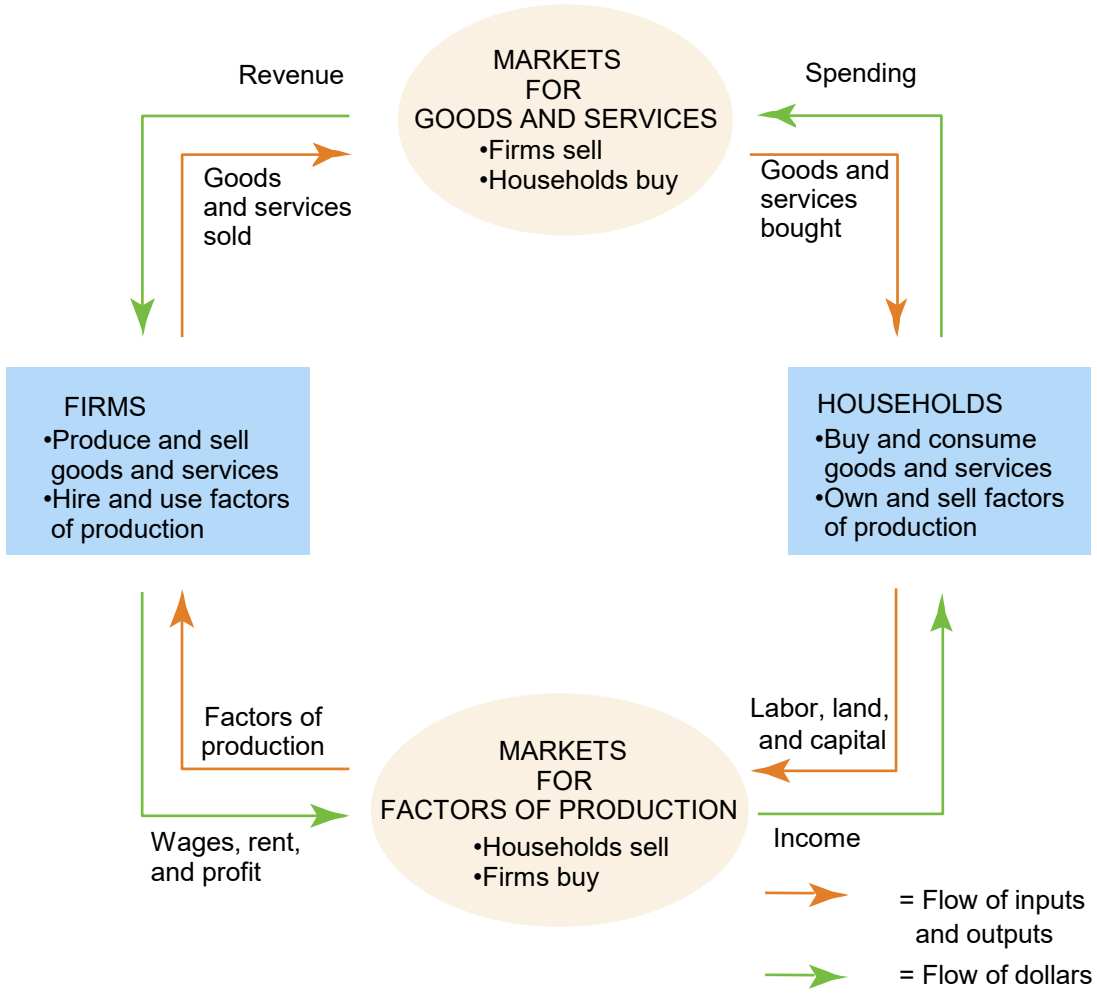
Which will be more useful on your next trip to London?

Understanding complex systems requires abstraction

Economic Models

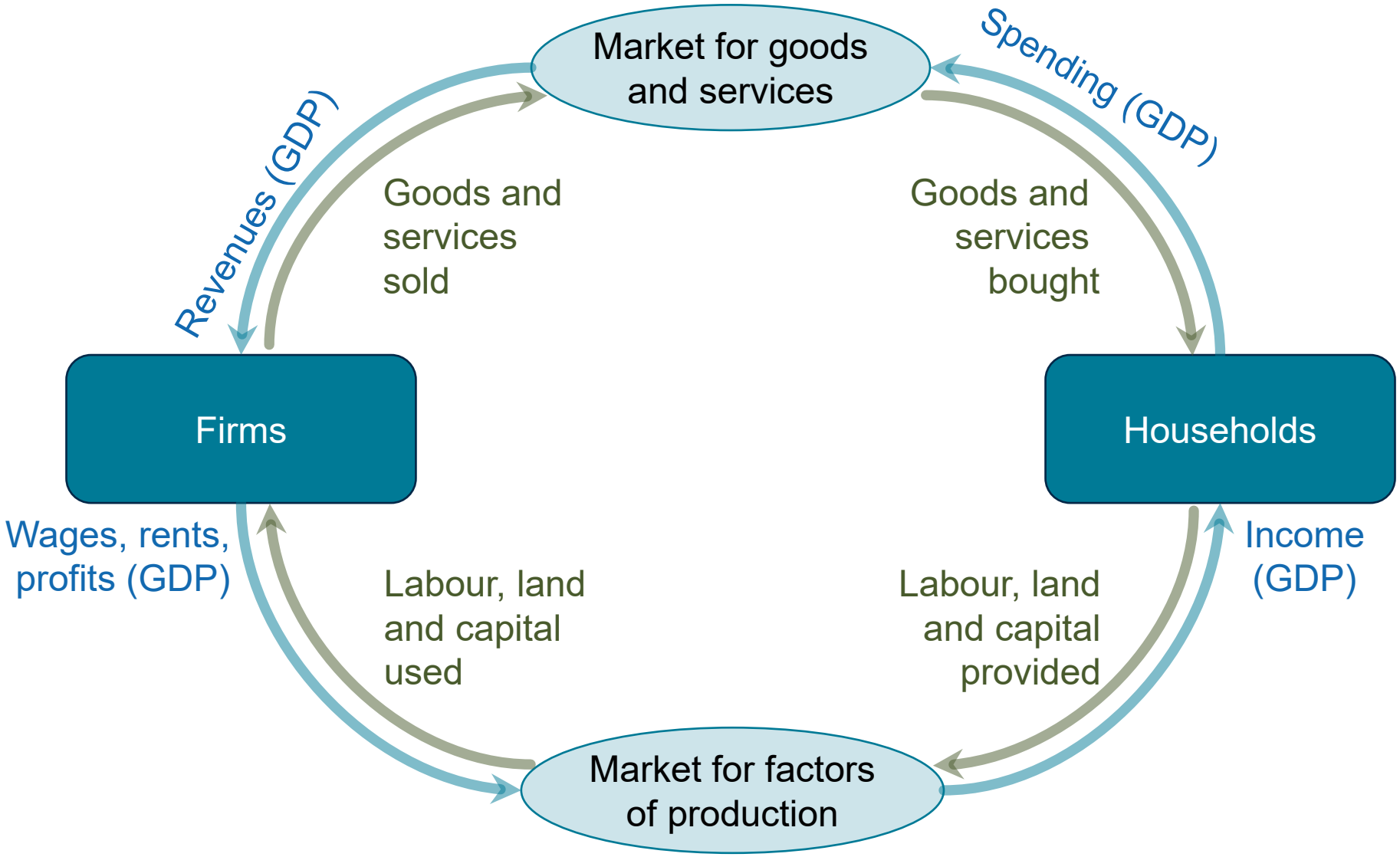
- Economists use models to simplify reality in order to improve our understanding of the world
- A model will contain a number of variables.
  - Definition of an endogenous variable:
    - a variable whose value is determined within the model.
  - Definition of an exogenous variable:
    - a variable whose value is determined outside the model.
  - Cause and effect
    - exogenous variables determine endogenous variables
  - Ceteris paribus meaning ‘other things equal’, is used to note when other factors that might affect outcomes are assumed to be constant
- Models are more unstable the longer the time period being considered and forecast
- Two of the most basic economic models are:
  - The Circular Flow Diagram
  - The Production Possibilities Frontier

Our First Model: The Circular-Flow Diagram



- The ***circular-flow diagram*** is a visual model of the economy that shows how dollars flow through markets among households and firms
  - Factors of Production
    - Inputs used to produce goods and services
    - Land, labour, and capital
  - Markets for Factors of Production
    - Households sell / Firms buy
  - Markets for Goods and Services
    - Firms sell / Households buy
  - Households
    - Buy and consume goods and services
    - Own and sell factors of production
  - Firms
    - Produce and sell goods and services
    - Hire and use factors of production

A circular-flow diagram of the economy



Financial flows in a circular-flow diagram in a more realistic economy

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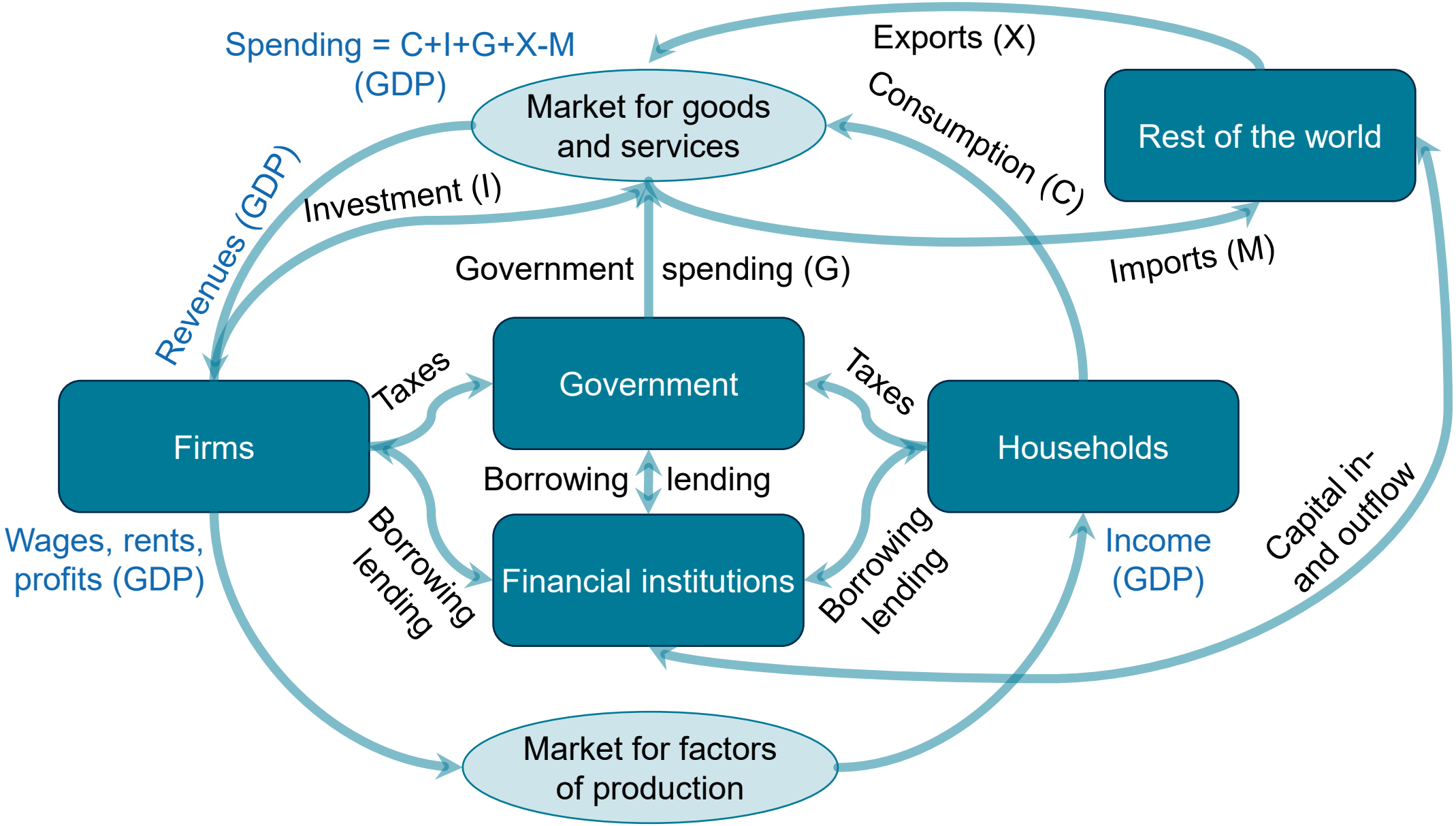
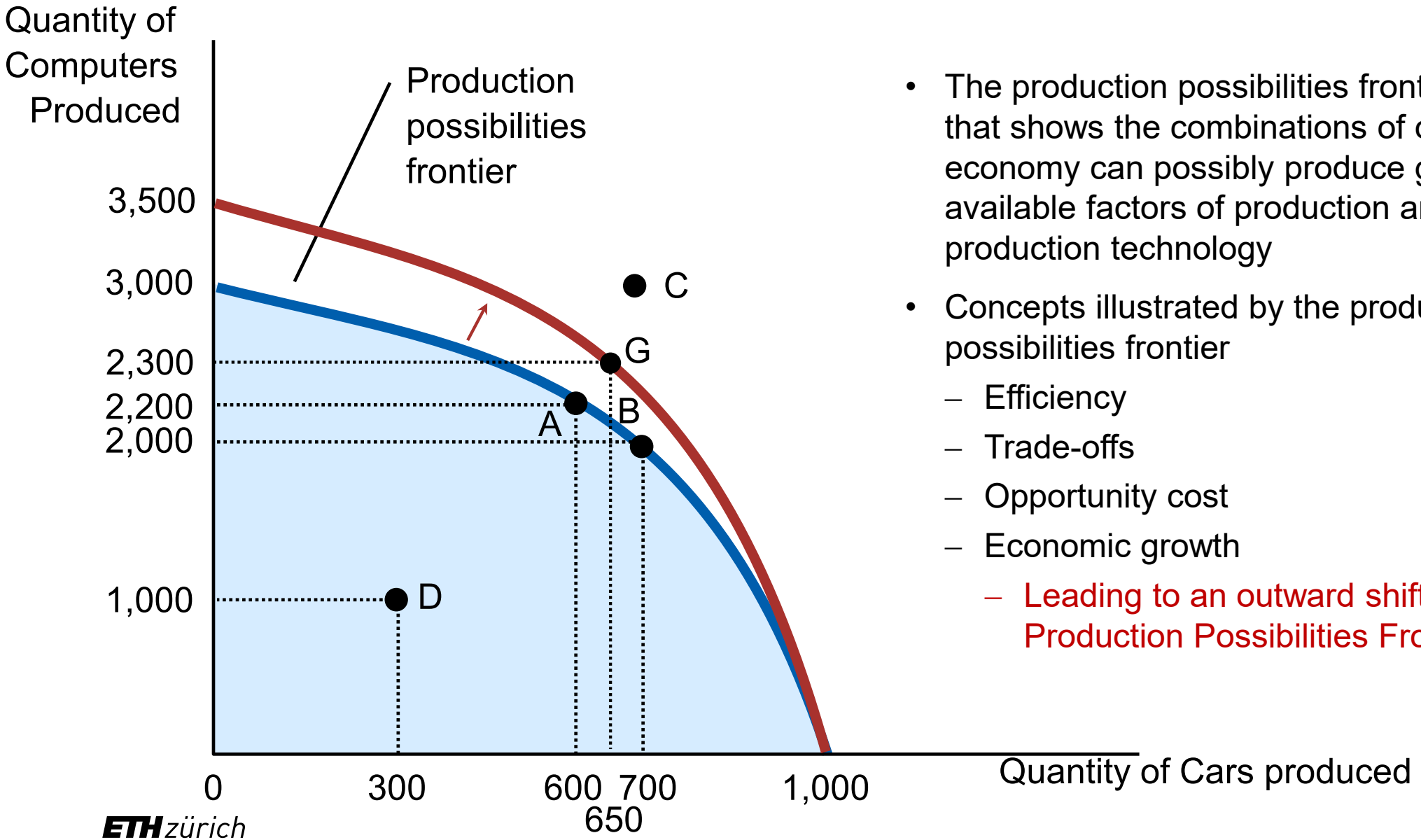


Our Second Model: The Production Possibilities Frontier



- The production possibilities frontier is a graph that shows the combinations of output that the economy can possibly produce given the available factors of production and the available production technology
- Concepts illustrated by the production possibilities frontier
  - Efficiency
  - Trade-offs
  - Opportunity cost
  - Economic growth
    - Leading to an outward shift of the Production Possibilities Frontier

THE ECONOMIST AS POLICY ADVISOR

- When economists are trying to explain the world, they are **scientists**
- When economists are trying to change the world, they are **policy advisors**

Positive versus Normative Analysis

- *Positive statements* are statements that attempt to describe the world as it is
  - Called descriptive analysis

- *Normative statements* are statements about how the world should be
  - Called prescriptive analysis

WHY ECONOMISTS DISAGREE

- They may disagree about the validity of alternative positive theories about how the world works
- They may have different values and, therefore, different normative views about what policy should try to accomplish

Do Economist Agree? (KOF-NZZ Economists Survey, December 2024)

https://kof.ethz.ch/en/surveys/experts-surveys/economists-surveys/econstat.html

- Guaranteed property rights and free competition are central to increasing a country's material well-being
  - 92.6% agree – 3.4% disagree
- Without publicly funded basic research, there is too little innovation and growth
  - 92.5% agree – 5.1% disagree
- Tariffs and import quotas reduce the material well-being in a country
  - 80.8% agree – 9% disagree
- Wage and/or price controls should be used as a tool to fight inflation
  - 6.8% agree – 78.4% disagree
- Taxing emissions is a better method to tackle pollution than setting limits
  - 77.8% agree – 10.7% disagree
- Climate-neutral growth will be possible with new technologies
  - 70.5% agree – 11.9% disagree
- Rent control reduces the quantity or quality of housing supply
  - 69.7% agree – 13.7% disagree
- The government expenditure share of GDP in Switzerland is too high
  - 13.1% agree – 65.3% disagree
- Disposable income in Switzerland should be distributed more evenly
  - 39.6% agree – 34.5% disagree

Summary (1)

- The principles of personal decision making are:
  - People face trade-offs.
  - The cost of something is what you give up to get it.
  - Rational people think at the margin.
  - People respond to incentives.
- The principles of economic interaction are:
  - Trade can make everyone better off.
  - Markets are usually a good way to organize economic activity.
  - Governments can sometimes improve market outcomes (if there is some market failure or if the market outcome is inequitable).
- The principles of the economy as a whole are:
  - A country's standard of living depends on its ability to produce goods and services.
  - Prices rise when the government prints too much money.
  - Society faces a short-run trade-off between inflation and unemployment.

Summary (2)

- Economics is divided into two subfields:
  - Microeconomics is the study of decision-making by households and firms in the marketplace.
  - Macroeconomics is the study of the forces and trends that affect the economy as a whole.
- A positive statement is an assertion about how the world is.
- A normative statement is an assertion about how the world ought to be.
- When economists make normative statements, they are acting more as policy advisors than scientists.
- Economists who advise policymakers offer conflicting advice either because of differences in scientific judgments or because of differences in values.
- At other times, economists are quite united in the advice they offer, but policymakers may choose to ignore it.